



Systems & Low-Level Software Engineer

Systems and low-level developer with a growing focus on data. Finishing among the top 10 students of my 42 Lausanne cohort. I combine the technical rigor of a self-taught engineer with the maturity of 10 years of professional experience. I build with full understanding, document with discipline and solve complex problems with patience. I am looking for a 4-to-6 month internship starting May 2026 with a company that values craftsmanship, intellectual honesty, and long-term thinking.

Alexandre Van der Plassche

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- 📍 Lausanne, Vaud
- 📄 B Permit EU/EFTA
- 🗣️ French (Native) | English (C1)
- 🌐 Portfolio: avdplassche.ch
- 🐙 Github : [avdplassche](https://github.com/avdplassche)

EXPERTISE

- Languages**
C/C++ · Bash · TypeScript · Python (in progress)
- Frameworks & Lib**
React · SFML · SDL3 · Crow
- Systems & Tools**
Linux (Debian, Arch) · Docker · Git · GDB · Valgrind · NeoVim · Insomnia · Doxygen
- Data Base**
SQLite · PostgreSQL

EDUCATION

- 42 Lausanne** 2024 - 2026
Software Development
- University of Montreal** 2013 - 2019
 - Bachelor in Economic Sciences
 - Major in Electroacoustic Studies

METHODOLOGY

Built and maintained a personal knowledge base using Zettelkasten and MOC structures covering CS fundamentals, adding theoretical foundations into 42's practice-first curriculum. Data flow and architecture mapping using Obsidian Canvas, Doxygen and draw.io prior to implementation.

PROJECTS

SoundCloud-like SPA (in progress) - React, Go (Gin), PostgreSQL, Docker

Collaborative full-stack application built as a team of 5, featuring audio streaming and an integrated musical sequencer. Contributed as front-end developer, product owner, and co-designed the relational database schema.

Webserv - C++, HTTP

Custom fully functional Nginx inspired HTTP server, built from scratch with C++98, handling concurrent connections, static file serving, and CGI execution.

MiniRT - C, Linear Algebra, Ray Tracing

Multithreaded ray tracing engine built from scratch to render 3D scenes. Capable of simulating complex lighting, shadows, and objects, showcasing mastery of linear algebra and rendering algorithms.

Systems & Simulation Independent Studies - C++, SFML / SDL3

A series of self-driven prototypes exploring real-time simulation and graphics programming. Includes a physics engine, a cellular automaton, and a simulation engine, exploring collision detection, vector math, and real-time rendering.

Nuzlocke Helper - C++, Crow, SQLite, React

Full-stack web application with a C++ REST backend (Crow framework), SQLite database, and React frontend. Delivers a high-performance utility for real-time tracking and session-based data storage.

EXPERIENCE

Independent Artist & Freelance Sound Designer 2015 - 2025

Released original music under the alias Vedou on established labels. Performed internationally (France, Canada, Belgium). Delivered sound design for short films. Managed all aspects of an independent creative practice - production, distribution, client relations, and project delivery.

Team Lead & Operations - Restaurant Industry 2015 - 2026

A decade of experience across multiple establishments, including temporary management responsibilities. Developed operational rigor, composure under pressure, and team coordination.

INTERESTS

Geopolitical History · Oenology · Tennis & Table Tennis
Creative Coding (generative art, TouchDesigner, p5.js)